

PRACTICAL 7

Task 1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;

void display_function()
{
    printf("Code segment address:  %p\n", (void *)display_function);
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;
    int *heap_var = (int *)malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    *heap_var = 40;

    display_function();
}
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C L
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ G

```
ruhi0404@Ruhitha:~/Operating_Systems$ nano prac7task1.c
ruhi0404@Ruhitha:~/Operating_Systems$ gcc prac7task1.c -o prac7task1
ruhi0404@Ruhitha:~/Operating_Systems$ gcc prac7task1.c -o prac7task1
ruhi0404@Ruhitha:~/Operating_Systems$ ./prac7task1
Code segment address:  0x603b1106d1c9
Global variable address: 0x603b11070010
Static variable address: 0x603b11070014
Heap variable address:  0x603b2de5e2a0
Stack variable address: 0x7ffec5013aec
ruhi0404@Ruhitha:~/Operating_Systems$ |
```

```

cat: /proc/500/maps: No such file or directory
ruhi0404@Ruhitha:~$ pgrep prac7task1
596
ruhi0404@Ruhitha:~$ cat /proc/596/maps
615032afe000-615032aff000 r--p 00000000 08:30 12101 /home/ruhi0404/Operating_Systems/prac7task1
615032aff000-615032b00000 r-xp 00001000 08:30 12101 /home/ruhi0404/Operating_Systems/prac7task1
615032b00000-615032b01000 r--p 00002000 08:30 12101 /home/ruhi0404/Operating_Systems/prac7task1
615032b01000-615032b02000 r--p 00002000 08:30 12101 /home/ruhi0404/Operating_Systems/prac7task1
615032b02000-615032b03000 rw-p 00003000 08:30 12101 /home/ruhi0404/Operating_Systems/prac7task1
615036172000-615036193000 rw-p 00000000 00:00 0 [heap]
7a24d3600000-7a24d3628000 r--p 00000000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7a24d3628000-7a24d37b0000 r-xp 00028000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7a24d37b0000-7a24d37ff000 r--p 001b0000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7a24d37ff000-7a24d3803000 r--p 001fe000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7a24d3803000-7a24d3805000 rw-p 00202000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7a24d3805000-7a24d3812000 rw-p 00000000 00:00 0
7a24d3946000-7a24d3949000 rw-p 00000000 00:00 0
7a24d394e000-7a24d3950000 rw-p 00000000 00:00 0
7a24d3950000-7a24d3954000 r--p 00000000 00:00 0 [vvar]
7a24d3954000-7a24d3956000 r--p 00000000 00:00 0 [vvar_vclock]
7a24d3956000-7a24d3958000 r-xp 00000000 00:00 0 [vdso]
7a24d3958000-7a24d3959000 r--p 00000000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a24d3959000-7a24d3984000 r-xp 00001000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a24d3984000-7a24d398e000 r--p 0002c000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a24d398e000-7a24d3990000 r--p 00036000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a24d3990000-7a24d3992000 rw-p 00038000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffe9947b000-7ffe9949c000 rw-p 00000000 00:00 0 [stack]
ruhi0404@Ruhitha:~$

```

Observation

The program successfully prints the memory addresses of code, global, static, heap, and stack variables. Each variable is stored in a specific memory segment of the Linux process address space. The code segment contains executable instructions, global and static variables are stored in the data segment, dynamically allocated memory is stored in the heap, and local variables are stored in the stack. The memory addresses may vary between executions due to Address Space Layout Randomization (ASLR).

Task 2: Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
GNU nano 7.2                                     prac7task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

int main()
{
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("Linux Process Virtual Memory Analysis\n");
    printf("-----\n");
    printf("Process ID (PID)          : %d\n", getpid());
    printf("Code address              : %p\n", (void *)main);
    printf("Global variable address   : %p\n", (void *)&global_var);
    printf("Static variable address   : %p\n", (void *)&static_var);
    printf("Heap variable address     : %p\n", (void *)heap_var);
    printf("Stack variable address    : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Enter to terminate.\n");

    ^G Help      ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C
```

```
ruhi0404@Ruhitha:~/Operating_Systems$ nano prac7task2.c
ruhi0404@Ruhitha:~/Operating_Systems$ gcc prac7task2.c -o prac7task2
ruhi0404@Ruhitha:~/Operating_Systems$ ./prac7task2
Linux Process Virtual Memory Analysis
-----
Process ID (PID)          : 622
Code address              : 0x60d88e7d5209
Global variable address   : 0x60d88e7d8010
Static variable address   : 0x60d88e7d8014
Heap variable address     : 0x60d898c822a0
Stack variable address    : 0x7fff5a9b777c

Process is running...
Press Enter to terminate.
```

```

7772dca00000-7772dca28000 r--p 00000000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7772dca28000-7772dccb0000 r-xp 00028000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7772dccb0000-7772dcbff000 r--p 001b0000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7772dcbff000-7772dcc03000 r--p 001fe000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7772dcc03000-7772dcc05000 rw-p 00202000 08:30 49808 /usr/lib/x86_64-linux-gnu/libc.so.6
7772dcc05000-7772dcc12000 rw-p 00000000 00:00 0
7772dcd65000-7772dcd68000 rw-p 00000000 00:00 0
7772dcd6d000-7772dcd6f000 rw-p 00000000 00:00 0
7772dcd6f000-7772dcd73000 r--p 00000000 00:00 0 [vvar]
7772dcd73000-7772dcd75000 r--p 00000000 00:00 0 [vvar_vclock]
7772dcd75000-7772dcd77000 r-xp 00000000 00:00 0 [vdso]
7772dcd77000-7772dcd78000 r--p 00000000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7772dcd78000-7772dcda3000 r-xp 00001000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7772dcda3000-7772dcdad000 r--p 0002c000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7772dcdad000-7772dcdaf000 r--p 00036000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7772dcdaf000-7772dccb1000 rw-p 00038000 08:30 49805 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff5a999000-7fff5a9ba000 rw-p 00000000 00:00 0 [stack]
ruhi0404@Ruhitha:~$ cat /proc/1234/maps | grep -E '\[heap\]|\[stack\]|prac7task2'
cat: /proc/1234/maps: No such file or directory
ruhi0404@Ruhitha:~$ cat /proc/622/maps | grep -E '\[heap\]|\[stack\]|prac7task2'
60d88e7d4000-60d88e7d5000 r--p 00000000 08:30 11932 /home/ruhi0404/Operating_Systems/prac7task2
60d88e7d5000-60d88e7d6000 r-xp 00001000 08:30 11932 /home/ruhi0404/Operating_Systems/prac7task2
60d88e7d6000-60d88e7d7000 r--p 00002000 08:30 11932 /home/ruhi0404/Operating_Systems/prac7task2
60d88e7d7000-60d88e7d8000 r--p 00002000 08:30 11932 /home/ruhi0404/Operating_Systems/prac7task2
60d88e7d8000-60d88e7d9000 rw-p 00003000 08:30 11932 /home/ruhi0404/Operating_Systems/prac7task2
60d898c82000-60d898ca3000 rw-p 00000000 00:00 0 [heap]
7fff5a999000-7fff5a9ba000 rw-p 00000000 00:00 0 [stack]
ruhi0404@Ruhitha:~$ █

```

Observation:

The `/proc/<PID>/maps` file was used to examine the virtual memory organization of a running Linux process. The process contains separate virtual memory regions for the executable code, data, heap, stack, and shared libraries. Each region has specific access permissions such as read, write, and execute. The memory mappings demonstrate that a Linux process uses a virtual address space divided into multiple regions. The virtual addresses are managed by the operating system and may vary between executions due to Address Space Layout Randomization (ASLR).